



**Met One Instruments, Inc.**

1600 NW Washington Blvd • Grants Pass, OR 97526 • (541) 471-7111 • www.metone.com

# Certificate of Calibration

*Model Spirant BAM*  
*Beta Attenuation Mass Monitor*

Calibrated in Grants Pass, OR USA

Spirant BAM Serial Number: **P10775**

Sales Order: \_\_\_\_\_

Calibration Performed By: **Tasey Graves**

Calibration Date: **02-14-2013**

Final Quality Inspection By: **Hengen**

Inspection Date: **3-21-13**

Configuration: ☐ PM<sub>10</sub> FEM ☐ PM<sub>2.5</sub> FEM ☒ PM<sub>2.5</sub> ☐ PM<sub>10-2.5</sub> Coarse FEM ☐ PM<sub>10</sub>EU ☐ PM<sub>2.5</sub>EU

Condition: ☒ New Unit ☐ Demonstration Unit ☐ Repair / Upgrade / Recalibration

The Spirant BAM is designated as a U.S. EPA Federal Equivalent Method (FEM) for PM<sub>10</sub> and PM<sub>2.5</sub> continuous particulate monitoring in accordance with 40 CFR Part 53, when equipped with the appropriate settings and accessories. Older revision units must be upgraded to the latest specs in order to be PM<sub>2.5</sub> FEM compatible. See the manual for details.

PM<sub>10</sub>: EQPM-0798-122

PM<sub>2.5</sub>: EQPM-0308-170

PM<sub>10-2.5</sub>: EQPM-0709-185

| Factory Calibration Settings |         |                                         |
|------------------------------|---------|-----------------------------------------|
| Name                         | Setting | Description                             |
| K                            | 0.991   | Concentration Slope Multiplier          |
| BKGD                         | 0.0030  | Concentration Offset, mg/m <sup>3</sup> |
| ABS                          | 0.797   | Span Membrane Mass, mg/cm <sup>2</sup>  |
| μsw                          | 0.312   | Absorption Coefficient                  |
| C <sub>v</sub>               | 0.967   | Flow Sensor Coefficient of Variability  |
| Q <sub>0</sub>               | 0.000   | Flow Sensor Zero Correction, SLPM       |

These settings are unit-specific calibration values which have been determined during dynamic testing, and should never be changed without specific instructions from Met One Instruments. See the operation manual. The BKGD background correction value may be field audited and altered.

| Other Unit Parameters                    |                 |
|------------------------------------------|-----------------|
| Spirant BAM Firmware:                    | 81236-02 v1.0.0 |
| Report Processor Firmware:               | N/A             |
| Panel PC Software:                       | N/A             |
| Beta Source Serial Number:               | UO 217          |
| Beta Detector Voltage, V <sub>DC</sub> : | -1025           |
| AC Power Supply Setting:                 | 220/50          |
| Concentration Stability Results          |                 |
| Hourly Sensitivity (σ):                  | 1.5             |
| Hourly Detection Limit (2σ):             | 3.0             |
| 24 Hour Detection Limit (2σ/5):          | 0.60            |

| Test and Calibration Standards |                  |               |                 |
|--------------------------------|------------------|---------------|-----------------|
| Standard Type                  | Mfg/Model        | Serial Number | Calibration Due |
| Air Flow Meter                 | BIOS DELTA CAL   | 001039        | 08-22-2013      |
| Digital Multimeter             | 189 MULTIMETER   | 82310054      | 06-14-2013      |
| Air Temperature                | MET ONE O83D-1-6 | N13974        | 11-07-2013      |
| Barometric Pressure            | MET ONE 091      | B5817         | 05-02-2013      |
| Relative Humidity              | MET ONE O83D-1-6 | N13974        | 11-07-2013      |
| Gravimetric Transfer Standard  | BAM - 1020       | N2404         | 11-14-2013      |

This instrument has been tested and calibrated to meet the manufacturer's published specifications at an ISO-9001 certified facility. The standards used for the calibration are on record and traceable to the National Institute of Standards and Technology (NIST), and have accuracies equal to or greater than the instrument being tested. The calibration system complies with MIL-STD-45662A. Complete test records for each unit are maintained by Met One Instruments, Inc. and are available upon request. The Spirant BAM mass measurement subsystem does not require any periodic recalibration except in cases of factory repairs to the beta measurement subsystem or its components.

Document No. Spirant-BAM-6101, Rev A



Met One  
Instruments

# Certificate of Compliance

The following product has passed a high voltage leakage test in accordance with Met One Instruments Test Procedure 7093-6100:

Model: Spirant BAM

S/N: P10775

Dated: 02-07-2013

Signed: 

**Met One Instruments, Inc.**  
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